

Author Response: TriPoseFusion: Robust Multi-View 3D Driver Pose Fusion with Outlier-Aware Canonicalization

To all reviewers. Prompted by R-LAYQ’s observation that PA-MPJPE exceeds MPJPE in Tab. 3 (with corrected results explicitly requested), we audited our full evaluation stack. The audit confirmed the reviewers’ concerns and found four defects: (i) a transposed rotation and norm-ratio scale in the Procrustes alignment; (ii) a zero-gradient gate-entropy regularizer (clamp after log; loss constant at $\ln 3$); (iii) a degenerate shoulder fallback collapsing whole frames — exactly as R-LAYQ inferred; (iv) uncalibrated constructed extrinsics in the triangulation reference, causing per-session metric-scale errors (median $2.4\times$). All are fixed (regression tests included); the reference is rebuilt by per-sequence self-calibration with anthropometric anchoring (reprojection $9.60 \rightarrow 3.31$ px; shoulder-width std $0.369 \rightarrow 0.015$ m; front-view utilization $8\% \rightarrow 45\%$). All numbers below use the corrected code and answer the reviewers’ explicit requests; fixes, calibration, and tests will be released.

1. R-LAYQ W1 — metric anomaly (corrected results requested). Cause: (i)+(iii). On synthetic rotated/scaled data the old code returns PA error 0.7–2.4 m where the truth is 0.04; the fix recovers the noise level. Corrected, same masks (a same-mask unaligned MPJPE is now also reported): canonicalized mean fusion 0.971/0.564 m on the original reference — the reversal is gone; 0.186/0.060 m on the calibrated reference.

2. R-19jf W1, R-LAYQ W3, R-xS2o M2 — reference reliability. Per-joint valid rates: head/shoulder–neck 1.00, hands 0.34; hands dominate the corrected error (0.316 m vs. torso 0.067, shoulder–neck 0.033). The revision reports hands separately, adds a torso-only protocol, and states the pipeline precisely (manual checks apply to the 2D inputs; triangulation is automatic DLT with a 40 px filter). In-cabin mocap being infeasible (as R-xS2o notes), independence comes from zero-shot evaluation against *Drive&Act’s official reference* (§4), breaking same-camera circularity; conclusions are stable across three reference versions.

3. R-xS2o W1–W2 (“rectify the collapse”), R-LAYQ W4, R-19jf m — gating collapse; claims narrowed. The collapse is real, but not forced by the entropy regularizer — defect (ii) made its gradient exactly zero. As requested, we diagnosed it: retraining with the fix, with $\lambda_{\text{gate}}=0$, and with a leave-one-view-out teacher all still yield 0.3333 gates — the collapse is architectural (under a median pseudo-teacher, uniform fusion is a trivial optimum). We narrow the claims accordingly: the working core is *canonicalization + uniform fusion* (corrected full model 0.1857/0.0593 $\equiv$ mean fusion 0.1858/0.0595, fold-0 val); temporal/velocity modules give no clean-input gain, as both reviewers noted. Addressing the requested rectification, corruption-augmented training of the same gate restores its function: a 1.9K-param

head assigns weight 0.037 to a zeroed view, degrading only +1–5% (mean fusion: +18%; weight exactly 0 for missing views); an 8K residual head improves clean MPJPE –14% (0.128 vs. 0.149, 5-fold cross-subject). Efficiency (R-LAYQ): 0.87M params, 3.3 MB, 6.2 ms/frame CPU.

4. R-xS2o requests 2–3 — Drive&Act (mandatory); learned baselines (requested). Zero-shot on Drive&Act (official reference, different rig, no training on it): canonicalize+mean **23.1 mm** PA-MPJPE, full model **23.7 mm** — within the 22.6–30.4 mm supervised-SOTA range the reviewer cites; DLT with the official calibration: 32.5 mm absolute; label-free per-rig adaptation of the 8K head: **18.6 mm** (–23%). The paper’s 0.412 m magnitude was the defect-(iv) scale artifact, not pose error; PCK@0.05/0.02 added. Baselines: the named methods (UPose3D, Learnable Triangulation, Faster VoxelPose) consume images/calibrated volumes and are inapplicable to our keypoint-only interface, whose purpose is precisely to avoid image-level fusion on embedded hardware. On the identical input we add the applicable learned family: confidence-weighted LT-style regression (1.9K), set-attention with uncertainty head (28K; Spearman 0.72), residual head (8K; best), original architecture (870K; worst) — 16 configurations, 5-fold.

5. R-19jf W2 — canonicalization mechanism. The gain is the removal of per-view depth/scale inconsistency, not the fixed rig: the same procedure transfers zero-shot to Drive&Act’s different geometry (23.1 mm), and raw→canonicalized single-view error drops on both datasets.

6. R-LAYQ W2 + minor items. Missing fallback rule (correctly inferred): temporally nearest valid shoulder axis within the clip, else a default lateral axis, with a parallel-case guard; pseudocode added to §3.3. Revision also adds: exact 52-joint mapping table, Group-KFold splits/durations, frame-rate-normalized derivatives, SAM3D architecture/domain details (R-xS2o M1).